ALLEN

ONLINE CLASSROOM CONTACT PROGRAMME

(Academic Session : 2025 - 2026)

Enthuse Course

PHASE : ALL

TARGET : Pre-Medical 2026

Test Type :**MAJOR**

Test Pattern :**NEET (UG)**

TEST DATE : 23-05-2026

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	**3**	**2**	**2**	**2**	**3**	**1**	**4**	**4**	**2**	**3**	**2**	**3**	**2**	**4**	**2**	**1**	**2**	**2**	**2**	**2**	**1**	**2**	**4**	**2**	**3**	**3**	**2**	**1**	**3**	**4**
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	**1**	**2**	**1**	**3**	**4**	**2**	**1**	**2**	**2**	**2**	**3**	**2**	**4**	**3**	**1**	**3**	**3**	**4**	**2**	**1**	**2**	**2**	**4**	**4**	**3**	**1**	**2**	**3**	**4**	**2**
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	**2**	**1**	**2**	**4**	**3**	**1**	**2**	**3**	**4**	**2**	**2**	**3**	**3**	**3**	**3**	**3**	**4**	**4**	**1**	**2**	**1**	**1**	**2**	**3**	**4**	**1**	**2**	**2**	**1**	**3**
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	**2**	**4**	**3**	**4**	**3**	**3**	**2**	**2**	**1**	**1**	**1**	**4**	**4**	**1**	**1**	**3**	**3**	**3**	**4**	**2**	**2**	**2**	**1**	**4**	**3**	**1**	**2**	**4**	**3**	**2**
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	**3**	**2**	**3**	**2**	**1**	**2**	**1**	**1**	**4**	**3**	**1**	**2**	**3**	**4**	**2**	**1**	**2**	**4**	**3**	**1**	**4**	**4**	**4**	**2**	**1**	**1**	**3**	**3**	**2**	**3**
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	**2**	**4**	**2**	**2**	**3**	**4**	**4**	**2**	**2**	**4**	**2**	**4**	**2**	**3**	**2**	**4**	**3**	**1**	**3**	**3**	**3**	**1**	**2**	**1**	**3**	**1**	**3**	**1**	**4**	**2**

HINT – SHEET

1. **Ans (3)**

EXPLANATION: The question asks us to evaluate two statements regarding an LCR series AC circuit and a purely resistive circuit. Statement I says current is maximum at resonance in a series LCR circuit. Statement II claims that current in a purely resistive circuit can never be less than that in a series LCR circuit of identical resistance when connected to the same voltage source.

CONCEPT : AC Circuits, Impedance (Z), and Resonance.

Solution :

Statement-I

$I_m = \dfrac{V_m}{\sqrt{R^2 + (X_L - X_C)^2}}$ at resonance $X_L = X_C$

Thus, $I_m = \dfrac{V_m}{R}$

$\because$ Impendence is minimum therefore I is maximum at resonance.

Statement-II

$I = \left(\dfrac{V}{R}\right)$ in purely resistive circuit.

2. **Ans (2)**

Question Explanation: A square wave current alternates between +0.5 A and −0.5 A; find the AC ammeter reading (RMS value).

Concept: RMS value of alternating current.

Solution: For a square wave, current magnitude is constant.

So, $I_{rms} = \sqrt{[(1/T) \times \int_0^T i^2(t)\, dt]}$

Since $i = \pm 0.5$ A, $i^2 = (0.5)^2 = 0.25$ A^2 always.

Hence, $I_{rms} = \sqrt{(0.25)} = 0.5$ A

Final Answer: $I_{rms} = 0.5$ A

3. **Ans (2)**

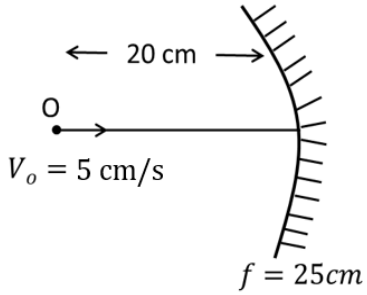
Question Explanation:

to calculate the velocity of image if object is moving with 5cm/s towards the mirror (concave)

Concept:

Velocity of image in spherical mirrors.

Solution:



u = −20cm

f = −25cm

$M_T = \dfrac{f}{f - u} = \dfrac{-25}{-25 - (-20)} = \dfrac{-25}{-5}$

$M_T = 5$

$\vec{V}_I = -M_T^2\, \vec{V}_O$

$\vec{V}_I = -(5)^2(5) = -125$cm/s

Final Answer: (2).

4. **Ans (2)**

Question Explanation:

Wavefronts are perpendicular to rays.

Concept: Wavefront Geometry.

Solution:

A wavefront is a surface where all points have the same phase.

For a light wave traveling along a specific axis, the wavefront must be a plane perpendicular to that direction of propagation.

Since the light is moving along the y-axis, the plane of the wavefront must be the xz-plane, which is represented by the equation y = constant.

Final Answer: Option(2)..

5. **Ans (3)**

Question Explanation :

• Goal: Find the phase difference ϕ required for destructive interference.

• Given: Wave equations $Y_1 = b\cos\omega t$ and $Y_2 = b\cos(\omega t + \phi)$.

Concept : Destructive Interference

Solution :

Conditions:

• $\phi = (2n + 1)\pi$ or $(2n + 1)180°$, where n = 0,1,2,…

• For n = 0 : $\phi = 180°$

Option Analysis:

• 0°,360°,720° : Even multiples of π, leading to constructive interference.

• 180° : Odd multiple of π, leading to destructive interference.

Final Answer : Option (3)

6. **Ans (1)**

Question Explanation:

Refractive index of a prism is $\mu = \cos ec(A/2)$. Find the minimum angle of deviation δ_m.

Concept:

At minimum deviation $r_1 = r_2 = A/2$ and $\mu = \sin((A + \delta_m)/2) / \sin(A/2)$.

Solution:

Given $\mu = \cos ec\,(A/2) = 1/\sin(A/2)$. Substitute into prism relation:

$$\frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\left(\frac{A}{2}\right)} = \frac{1}{\sin\left(\frac{A}{2}\right)}$$

Hence $\sin\left(\frac{A + \delta_m}{2}\right) = 1$. Therefore $\frac{A + \delta_m}{2} = 90°$ (principal value), so $\delta_m = 180° - A$. (This is valid for $0 < A < 180°$,giving a positive deviation.)

Final Answer: Option (1)

7. **Ans (4)**

Explain Question:

Pick Out the incorrect statement from the given options.

Concept: Forward biasing of a p-n junction, depletion region and charge carrier movement.

Solution:

In forward biasing width of depletion layer decreases and free electrons move from n-side to p-side and holes move from p-side to n-side means charge carries move towards the junction

So, option 4 is incorrect statement.

Final Answer: Option (4)

8. **Ans (4)**

Explain Question:

Analyze $V_{AB}(t)$ across resistor R in a sinusoidal input circuit with two diodes and unequal resistors R_1 and R_2.

Concept:

Current direction changes with input cycle; unequal resistors cause different voltage drops across R.

Solution:

In the positive half cycle, diode D_2 is forward biased and diode D_1 is reverse biased. In the negative half cycle, diode D_1 is forward biased and D_2 is reverse biased and direction of current is opposite through R for positive and negative half cycles. Hence, output is not rectified. Since R_1 and R_2 are different, hence the peaks during positive half and negative half of the input signal will be different.

Final Answer : 4

9. **Ans (2)**

Question Explanation :

A circular plate of diameter a is in contact with a square plate of edge a. We need to find the center of mass of the composite system.

Concept Based :

This problem involves finding the center of mass of a composite system made of two shapes: a circle and a square. The center of mass is determined by the masses and positions of the objects.

Formula Used :

The center of mass (CM) for two objects is given by:

$$x_{CM} = \frac{M_{circle}x_{circle} + M_{square}x_{square}}{M_{circle} + M_{square}}$$

where M is the mass and x is the position of the center of mass of each object.

Calculation :

Given that the density and thickness are the same for circular and square plates, the masses of the two plates are proportional to their surface areas.

• Mass of the circle:

$$M_{circle} \propto \pi\left(\frac{a}{2}\right)^2 = \frac{\pi a^2}{4}$$

• Mass of the square:

$$M_{square} \propto a^2$$

Since a^2 is greater than $\frac{\pi a^2}{4}$, the square plate has more mass.

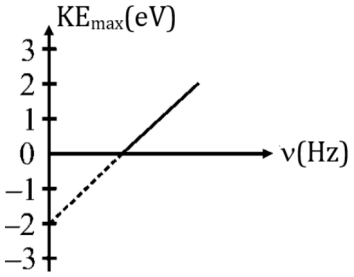
Thus, the center of mass of the system will be closer to the square plate, inside the square plate.

Final Answer: inside the square plate

Hence, the correct answer is option (2).

10. **Ans (3)**

Question Explanation : For a Photo-Electric Effect (PEE) experiment, a plot between the maximum kinetic energy of photo-electrons (KE_{max}) and frequency of the incident photon (ν_{Ph}) is a straight line and shown below :



We need to find the work-function of the illuminated metal surface.

Concept : Einstein's Photo-Electric Equation

Solution : According to the Einstein's Photo-Electric Equation -

$\phi_{Ph} = \phi_0 + KE_{max}$

$\Rightarrow h\upsilon_{Ph} = \phi_0 + KE_{max}$

$\Rightarrow KE_{max} = h\upsilon_{Ph} - \phi_0$ .....(i)

The equation of a straight line -

$y = mx + C$ .....(ii)

On comparing equations no. (i) & (ii), taking $y = KE_{max}$ & $x = \nu_{Ph}$, we get -

y intercept $\Rightarrow C = -\phi_0 =$ – Work Function of metal

Work Function of metal = $\phi_0 = - C = - (- 2\text{ eV}) = 2\text{ eV}$

Final Answer : Option (3)

11. **Ans (2)**

Question explanation:

Identify which of the given nuclear reactions does not occur within the standard proton-proton (p-p) chain reaction cycle.

Concept:

Stellar nucleosynthesis via the proton-proton chain reaction in stars like the Sun.

Solution:

The proton-proton cycle consists of three primary stages that fuse hydrogen into helium.

Stage 1 involves the fusion of two protons to form deuterium, a positron, and a neutrino.

$${}_1H^1 + {}_1H^1 \rightarrow {}_1H^2 + \beta^+ + V + Q$$

Stage 2 involves a deuterium nucleus fusing with a proton to form helium-3.

$${}_1H^2 + {}_1H^1 \rightarrow {}_2He^3 + Q$$

Stage 3 (p-p I branch) involves two helium-3 nuclei fusing to form helium-4 and two protons.

$${}_2He^3 + {}_2He^3 \rightarrow {}_2He^4 + 2({}_1H^1) + Q$$

Option 2 shows the fusion of two deuterium nuclei to form helium-3 and a neutron.

$${}_1H^2 + {}_1H^2 \rightarrow {}_2He^3 + n^1 + Q$$

This specific reaction is part of the deuterium-deuterium (D-D) fusion process, not the standard solar proton-proton cycle.

The proton-proton cycle does not produce free neutrons in its primary branches.

Final answer: (2)

12. **Ans (3)**

Concept:

Electromagnetic Spectrum - Frequency Order

Explanation:

The electromagnetic spectrum follows this decreasing frequency order:

-rays > X-rays > Ultraviolet > Visible > Infrared (Heat Rays) > Microwaves > Radio Waves

From this, we observe:

γ-rays > X-rays > Heat Rays (Infrared) > Radio Waves

Final Answer:

Option (3)

13. **Ans (2)**

Explain the Question : The problem involves comparing the velocities v_1 & v_2 of photo-electrons emitted from identical photo-cathodes when illuminated by the light of two different frequencies f_1 & f_2 , both above the threshold frequency f_0.

Concept : This question is based on Photo-Electric effect : Einstein's Photo-Electric equation.

Solution : Einstein's Photo-Electric equation-

$$\phi_{Ph} = \phi_0 + KE_{max}$$

$$hf = hf_0 + \frac{1}{2}mv^2$$

For frequency f_1 :

$$hf_1 = hf_0 + \frac{1}{2}mv_1^2$$

$$\frac{1}{2}mv_1^2 = h(f_1 - f_0) \quad \text{------(1)}$$

For frequency f_2 :

$$hf_2 = hf_0 + \frac{1}{2}mv_2^2$$

$$\frac{1}{2}mv_2^2 = h(f_2 - f_0) \quad \text{------(2)}$$

Subtracting (2) from (1) :

$$\frac{1}{2}m\left(v_1^2 - v_2^2\right) = h(f_1 - f_0) - h(f_2 - f_0)$$

$$\frac{1}{2}m\left(v_1^2 - v_2^2\right) = h(f_1 - f_2)$$

$$\left(v_1^2 - v_2^2\right) = \frac{2h}{m}(f_1 - f_2)$$

Final answer : 2

14. **Ans (4)**

Explanation Question: Velocity of particle is given and we have to represent quantities in order

Concept: This question is based on dimension analysis

Solution:

$[b] = \frac{LT^{-1}}{T} \Rightarrow$ acceleration

$[c] = LT^{-1}T = L \Rightarrow$ Distance

$[d] = T \Rightarrow$ Time

$[a] = \frac{L^2T^{-2}}{LT^{-2}} = L \Rightarrow$ Distance

Final Answer

The correct option is (4)

15. **Ans (2)**

Question Explanation: We have to find percentage error in x

Concept : This question is based on error propagation.

Solution :

$a = 8 \pm 0.04, b = 6 \pm 0.02$

$x = a - b$

$\Delta x = \Delta a + \Delta b$

$x \pm \Delta x = (a - b) \pm (\Delta a + \Delta b)$

$x \pm \Delta x = (8 - 6) \pm (0.04 + 0.02) = 2 \pm 0.06$

percentage error in 'x' :

$\frac{\Delta x}{x} \times 100 = \frac{0.06}{2} \times 100 = 3\%$

Final answer : Option (2)

16. **Ans (1)**

Question Explanation: Calculate the physical horizontal distance between two coins dropped sequentially at 1-second intervals from a uniformly accelerating car.

Concept Involved: Kinematics of a projectile released from a moving platform.

Solution:

Speed of Car at t = 3 is 6 m/s

so distance travelled by coin = 6 × 1 = 6m

speed of Car at t = 4 is 8 m/s

so distance travelled by coin = 8 × 1 = 8m

distance travelled by car during 3 to 4 sec

$S = 6 \times 1 + \frac{1}{2}2(1)^2 = 7m$

so distance between coin

$(7 + 8) - (6) = 9m$

Final Answer: (1)

17. **Ans (2)**

Question Explanation: Find the time taken for two uniformly accelerating trains moving in opposite directions to fully pass one another.

Concept Involved: Relative motion with constant acceleration in a linear track.

Solution :

As $v^2 = u^2 + 2as \Rightarrow (2u)^2 = u^2 + 2as = 3u^2$

Now, after covering an additional distance s, if velocity becomes u, then,

$v^2 = u^2 + 2a(2s) = u^2 + 4as = u^2 + 6u^2 = 7u^2$

$\therefore \quad u = \sqrt{7}u$

$\therefore \quad t = 10$ sec (Taking +ve value)

Final Answer: (2)

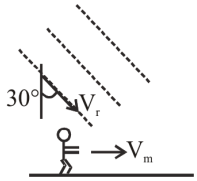
18. **Ans (2)**

Question Explanation: Calculate the absolute velocity of rain relative to the ground using changing apparent angles observed by a moving runner.

Concept Involved: Relative motion of rain with respect to a moving observer in a vertical plane.

Solution :

Given $V_r \cos 30° = 10$ m/s



$$V_r = \frac{20}{\sqrt{3}} \text{ m/s}$$

Final Answer: (2)

19. **Ans (2)**

Explain Question:

Find the tension at the top of a heavy rope connecting two blocks under an upward force.

Concept:

Newton's Second Law for a composite system with a non-massless rope.

Solution:

Total mass $M = 7 + 4 + 5 = 16$ kg

Net upward force $F_{net} = 200 - (16 \times 10) = 40$ N

Acceleration $a = \frac{40}{16} = 2.5 \text{ m/s}^2$

To find tension at point P, consider the system below P (rope + 5 kg block).

Mass below P is $m' = 4 + 5 = 9$ kg

$T_P - m' g = m' a$

$T_P = 9 \times (10 + 2.5) = 112.5$ N

Final answer: Option (2)

20. **Ans (2)**

Question Explanation: Find the mechanical work done by a two-dimensional variable force field on a particle traveling along a straight path.

Concept Involved: Work done as a line integral of a variable position-dependent force vector.

Solution :

$y = mx + c$

$$w = \int_0^5 2 \times dx + \int_0^5 y^2 dy = (x^2)_0^5 + \left(\frac{y^3}{3}\right)_0^5$$

$$25 + \frac{125}{3} = \frac{200}{3} = 66.7 \text{ J}$$

Final Answer: (2)

21. **Ans (1)**

Explain question : Find momentum of particle

Concept: This question is based on centripetal force = radiial force

Solution:

$$a_c = \frac{v^2}{r} = \frac{2}{r^3}$$

$$v^2 = \frac{2}{r^3} \times r \Rightarrow v = \sqrt{\frac{2}{r^2}} = \frac{\sqrt{2}}{r}$$

so momentum $p = mv = \frac{m\sqrt{2}}{r} = \frac{\sqrt{2}\, m}{r}$

Final Answer: (1) $\frac{\sqrt{2}m}{r}$

22. **Ans (2)**

Question Explanation :

A circular plate of diameter a is in contact with a square plate of edge a. We need to find the center of mass of the composite system.

Concept Based :

This problem involves finding the center of mass of a composite system made of two shapes: a circle and a square. The center of mass is determined by the masses and positions of the objects.

Formula Used :

The center of mass (CM) for two objects is given by:

$$x_{CM} = \frac{M_{circle}x_{circle} + M_{square}x_{square}}{M_{circle} + M_{square}}$$

where M is the mass and x is the position of the center of mass of each object.

Calculation :

Given that the density and thickness are the same for circular and square plates, the masses of the two plates are proportional to their surface areas.

• Mass of the circle:

$$M_{circle} \propto \pi\left(\frac{a}{2}\right)^2 = \frac{\pi a^2}{4}$$

• Mass of the square:

$$M_{square} \propto a^2$$

Since a^2 is greater than $\frac{\pi a^2}{4}$, the square plate has more mass.

Thus, the center of mass of the system will be closer to the square plate, inside the square plate.

Final Answer: inside the square plate

Hence, the correct answer is option (2).

23. **Ans (4)**

Question Explanation: Calculating the moment of inertia for a composite body.

Concept: Moment of Inertia and Parallel Axis Theorem.

Solution: The total moment of inertia is the sum of the moments of inertia of the two individual rods about the axis at O. For the horizontal rod, the axis is at its end. For the vertical rod, the parallel axis theorem ($I = I_{cm} + md^2$) must be used.

$$\left(\frac{m\ell^2}{3}\right) + \left(\frac{m\ell^2}{12} + mx^2\right)$$

where $x^2 = \ell^2 + \ell^2/4$

Moment of inertia about its CM is $I_{cm} = \frac{1}{12} ml^2$.

- $I_2 = I_{cm} + md^2 = 121\, ml^2 + m(45l^2) = 1216\, ml^2 = 34\, ml^2$

Total Moment of Inertia:

$I_{total} = I_1 + I_2 = 31\, ml^2 + 34\, ml^2 = 35\, ml^2$.

Final Answer: The correct option is (4)

- **Final Answer:** $\frac{5}{3}m\ell^2$

24. **Ans (2)**

Explain the question : In this question, a wheel-an-axle system is shown, and we are asked to determine the condition for the system to remain in static equilibrium.

Concept: This question is based on the conditions of static equilibrium.

Solution: Using torques balance about center of the wheel.

$am_1g = bm_2g$

$am_1 = bm_2$

Final Ans: 2

25. **Ans (3)**

Question Explanation: Compute the maximal vertical peak height reached by an upward launched ball using fractions of earth's absolute escape velocity.

Concept Involved: Universal law of gravitational potential energy conservation over extreme macroscopic displacements.

Solution :

COME

$$-\frac{GMm}{R} + \frac{1}{2}m(nVe)^2 = -\frac{GMm}{R+h}$$

$$\frac{1}{2}n^2 \times \frac{2GM}{R} = \frac{GM}{R} - \frac{GM}{R+h}$$

$$n^2\left(\frac{GM}{R}\right) = \frac{GM(R+h-R)}{R(R+h)}$$

$$n^2R + n^2h = h \Rightarrow (1-n^2)h = n^2R$$

$$h = \frac{n^2R}{1-n^2}$$

Final Answer: (3)

26. **Ans (3)**

Question Explanation: Calculate the final terminal velocity, when eight identical microscopic water droplets coalesce into one larger spherical entity.

Concept Involved: The terminal velocity (v) of a spherical body falling through a viscous fluid is given by Stokes' Law terminal velocity is directly proportional to the square of the radius.

Solution :

We know that,

$$v_{terminal} \propto r^2$$

$$\therefore \frac{v_1}{v_2} = \frac{r_1^2}{r_2^2}$$

when 8 drops combine, we have

$$8 \times \frac{4}{3}\pi r_1^3 = \frac{4}{3}\pi r_2^3$$

$$\Rightarrow r_2 = 2r_1$$

$$\therefore \frac{16}{v_2} = \frac{4}{16} \Rightarrow v_2 = 64 \text{ cm/sec}$$

Final Answer: (3)

27. **Ans (2)**

Explain the Question : Two wires of the same material, but with different lengths and radii, are stretched with different forces. We need to find the ratio of their elongations.

Concept : This question is based on elongation depends on force, length, cross-sectional area, and Young's modulus. It is directly proportional to force and length, and inversely proportional to area.

Formula :

$$\Delta L = \frac{FL}{AY}$$

Solution :

$$\Delta L_1 = \frac{F\ell}{\pi R^2 Y}$$

For wire 2 :

$$\Delta L_2 = \frac{2F(2\ell)}{\pi(2R)^2 Y} = \frac{4F\ell}{\pi(4R^2)Y} = \frac{F\ell}{\pi R^2 Y}$$

Thus : $\Delta L_1 = \Delta L_2$. so the ratio is 1 : 1.

Final answer : 2

28. **Ans (1)**

Question Explanation: Find heat transfer during expansion if temperature of the gas increases by 50K. During expansion, volume and temperature of gas vary such that $\frac{V}{T^2}$ = constant.

Concept Involved: First Law of Thermodynamics.

Solution :

$$VT^{-2} = \text{constant}$$

$$\Rightarrow PV^{\frac{1}{2}} = \text{constant}$$

$$Q = \frac{nR\Delta T}{\gamma - 1} + \frac{nR\Delta T}{1-x}$$

$$= \frac{nR50}{\frac{7}{5}-1} + \frac{nR50}{1-\frac{1}{2}}$$

$$= 225R$$

Final Answer: (1)

29. Ans (3)

Question Explanation : Find the total heat absorbed by a system undergoing a cyclic process represented by a circle on a V-P (Volume-Pressure) diagram, where the volume ranges from 300 m^3 to 500 m^3 and the pressure ranges from 400 N/m^2 to 600 N/m^2.

Concept: First Law of Thermodynamics for Cyclic Processes.

Formula : Q = W = Area of the ellipse = πab

Solution: For any cyclic process, the system returns to its initial state, meaning the total change in internal energy is zero ($\Delta U = 0$). According to the first law of thermodynamics ($Q = \Delta U + W$), the net heat exchanged equals the net work done by the system ($Q = W$).

Q = W =Area of the closed loop = π(semi-major axis a)(semi-minor axis b)

Semi-axis along P (horizontal): $a = \frac{600-400}{2} = 100\ N/m^2$

Semi-axis along V (vertical): $b = \frac{500-300}{2} = 100\ m^3$

W =π 100 x100 = 10000 x 3.14 = 31400 J

Final answer: Option (3)

30. Ans (4)

Question Explanation:

Determine the work done along path ib$\int$ given information about path ia$\int$.

Concept:

Internal energy is a state function; dU is the same for all paths between two points.

Solution:

For path ia$\int$: dU = dQ -dW =50-20=30 J

Since i and $\int$ are the same for path ib$\int$, $dU_{ib\int}$ = 30 J

For path ib$\int$: dQ = 36 J

dW=dQ-ΔU=36-30

dW=6J

Final answer: option (4)

31. Ans (1)

1. Explain question We must find the period of a pendulum when the lift descends with acceleration a = g/3. **2. Concept** Pendulum in accelerating frame

3. Solution Effective gravity:

$g_{eff} = g - a = g - \frac{g}{3} = \frac{2g}{3}$ Time period:

$T = 2\pi\sqrt{\frac{1}{g_{eff}}} = 2\pi\sqrt{\frac{1}{2g/3}} = 2\pi\sqrt{\frac{3l}{2g}}$

4. Final answer Option 1

32. Ans (2)

Question Explanation: A particle undergoing simple harmonic motion (SHM) with time period T. We have to determine the time taken for $\left(\frac{7}{8}\right)$ oscillation

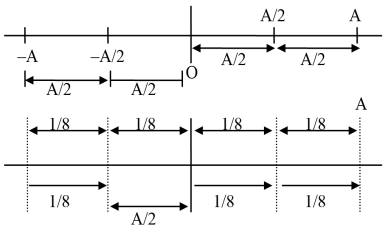
Concept: Time period in SHM.

Solution: Based on time taken by particle for motion of various distances along the line performing SHM.

Distance covered in 1 oscillation = 4A (A $\longrightarrow$ amplitude)

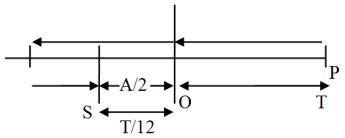
For $\frac{1}{8}$ oscillation, distance covered = $\frac{1}{8} \times 4A = \frac{A}{2}$

For $\frac{7}{8}$ oscillation, distance covered = $\frac{7}{8} \times 4A = 7 \times \frac{A}{2}$



After $\frac{7}{8}$ oscillation it is at position $-\frac{A}{2}$

Now from this position time taken by the particle to again reach at mean position is $\frac{T}{12}$



$\therefore$ Time taken to complete $\frac{7}{8}$oscillation is

$T - \frac{T}{12} = \boxed{\frac{11T}{12}}$

Final Answer: Option (2)

33. Ans (1)

Question Explanation: Find frequency and velocity of wave

Concept Involved: $\omega = 2\pi n$ and velocity $= n\lambda$ and $k = \frac{2\pi}{\lambda}$

Solution :

$$y = 4\sin 2\pi\left[\frac{t}{0.02} - \frac{x}{100}\right]$$

comparing with

$$y = A\sin 2\pi\left[\frac{t}{T} - \frac{x}{\lambda}\right]$$

$T = 0.02$ sec, $\lambda = 100$ cm

$$f = \frac{1}{T} = \frac{1}{0.02} = \frac{100}{2} = 50 \text{ Hz}$$

$$v = f\lambda = 50 \times \frac{100}{100} = 50 \text{ m/s}$$

Final Answer: (1)

34. Ans (3)

Question Explanation: Find the frequency of tunning fork.

Concept Involved: tunning fork.

Solution :

Tuning fork 'n' Hz

250 Hz → 5 beats/sec → Tuning fork 'n' Hz → 2 beats/sec → 257 Hz

Therefore, frequency of tuning fork = 255 Hz

Final Answer: (3)

35. Ans (4)

Question Explanation: Determine the central electrostatic scalar potential at the shared interior locus of concentric uniform conductive spherical boundaries.

Concept Involved: Electrostatic potential

Solution :

$$q_1 + q_2 = Q \text{ and } \frac{q_1}{4\pi r^2} = \frac{q_2}{4\pi R^2} \text{ (given)}$$

$$q_1 = \frac{Qr^2}{R^2 + r^2} \text{ and } q_2 = \frac{QR^2}{R^2 + r^2}$$

Potential at common centre

$$\frac{1}{4\pi\varepsilon_0}\left[\frac{Qr^2}{(R^2 + r^2)r} + \frac{QR^2}{(R^2 + r^2)R}\right] = \frac{Q(R + r)}{4\pi\varepsilon_0(R^2 + r^2)}$$

Final Answer: (4)

36. Ans (2)

Question Explanation: To find the correct statement regarding Gauss' law

Concept Involved: Gauss' law

Solution :

Solution: The standard mathematical expression shows the electric flux over a closed interface depends exclusively on enclosed interior charge. However, the vector field value E at any point is an absolute consequence of all spatial charges located both inside and outside the defined boundary. Thus statement 2 is factually sound.

Final Answer: (2)

37. Ans (1)

Question Explanation: To find the discharging time in terms of time constant.

Concept Involved: discharging of capacitor.

Solution :

$$\frac{1}{2}\frac{Q^2}{C} = \frac{1}{100}\left(\frac{1}{2}\frac{Q_0^2}{C}\right) \Rightarrow Q = \frac{Q_0}{10}$$

$$\frac{Q_0}{10} = Q_0 e^{-t/\tau} \Rightarrow \ell n(10)^{-1} = \frac{-t}{\tau}$$

$$\frac{t}{\tau} = 2.303$$

Final Answer: (1)

38. Ans (2)

Explain question:

Predicting brilliance changes in a circuit when one component fails.

Concept:

Series-Parallel Bulb Circuits.

Solution:

Lamps L_2 and L_3 are in parallel, and their combination is in series with L_1

- If L_3 fuses, the total circuit resistance increases (L_2 alone vs $L_2 || L_3$).
- Higher total resistance decreases main circuit current through L_1, making dimmer.
- However, more voltage becomes available of the remaining L_2 branch, making it brighter.

Final answer: option (2)

39. **Ans (2)**

EXPLANATION: To find the reading of ammeter in the circuit given.

Concept : Basic circuit concepts of current.

Solution :

Reading of $A_1 = \frac{12V}{(20\Omega/5)} = 3A$

Reading of $A_2 = \frac{12V}{(20\ \Omega/2)} = 1.2\ A$

Reading of $A_3 = \frac{12V}{20\ \Omega} = 0.6\ A$

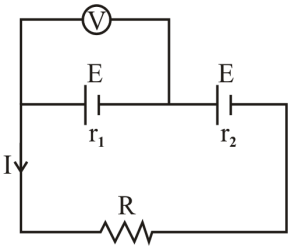
Final Answer: (2)

40. **Ans (2)**

Question Explanation: Formulate the relationship among system resistances under the specific constraint

Concept Involved: Internal source terminal voltage drop characteristics under complete loop current conduction regimes.

Solution :



Here voltmeter is ideal

so $I = \frac{2E}{R + r_1 + r_2}$

so $V = E - Ir_1 = 0$

$E = \frac{2Er_1}{R + r_1 + r_2} \Rightarrow R = r_1 - r_2$

Final Answer: (2)

41. **Ans (3)**

EXPLANATION: Two long parallel wires separated by a distance d carry currents I_1 and I_2 in the same direction. We need to identify the nature and magnitude of the magnetic force per unit length acting between them.

Concept : Force between two parallel current-carrying conductors.

Solution :

The magnetic induction midway is

$B_1 = \frac{\mu_0(i_1 - i_2)}{2\pi r} = 10\mu T$ when currents align. In reversal, fields superimpose to yield

$B_2 = \frac{\mu_0(i_1 + i_2)}{2\pi r} = 30\mu T$. Taking ratios:

$\frac{i_1 + i_2}{i_1 - i_2} = \frac{30}{10} = 3 \Rightarrow i_1 + i_2 = 3i_1 - 3i_2 \Rightarrow 2i_1 = 4i_2$

$\Rightarrow \frac{i_1}{i_2} = 2.$

Final Answer: (3)

42. **Ans (2)**

EXPLANATION: To find the maximum velocity of projection for which the particle does not hit the plate

Concept : charged particles moving in uniform magnetic fields.

Solution :

$Bqv \sin\frac{\pi}{2} = \frac{mv^2}{r}$

or, $r = \frac{mv}{Bq}$

The particle does not hit the plate if,

$r \le d$

or $\frac{mv}{Bq} \leqslant d$ or $v \leqslant \frac{Bqd}{m}$

$\therefore v_{max} = \frac{Bqd}{m}$

Final Answer: (2)

43. Ans (4)

EXPLANATION: A proton (p), a deuteron (d), and an alpha-particle, each starting from rest, are accelerated through the same potential difference V and enter a perpendicular uniform magnetic field. We need to find the ratio of their kinetic energies.

Concept : Electrostatic acceleration and work done on charged particles.

Solution :

Let the charge of a proton be $q_p = e$.

A deuteron has charge $q_d = e$.

An alpha-particle has charge $q_\alpha = 2e$.

Since the accelerating potential V is identical for all three particles, K.E. is proportional to q.

Ratio of Kinetic Energies = $q_p : q_d : q_\alpha = e : e : 2e = 1 : 1 : 2$.

44. Ans (3)

EXPLANATION: A wire carrying current i is shaped according to the equation $y = a \times \sin(\pi \times x / L)$ from $0 \le x \le 2L$ and is placed inside a uniform magnetic field B. We need to compute the net magnetic force acting on the wire.

Concept : Magnetic force on a curved wire in a uniform field depends only on its straight effective length vector (L_{eff}) connecting the initial and final endpoints.

Solution :

Effective length of the wire is 2L

$F = i B (2L)$

45. Ans (1)

EXPLANATION: A flat coil of area A is positioned in a magnetic field with its plane parallel to the field lines. We need to determine average emf induced in the coil.

Concept : Faraday's Law of Electromagnetic Induction

Solution :

$$e = -\frac{d\phi}{dt} = \frac{-3B_0A_0}{t}$$

Final Answer: (1)

46. Ans (3)

Explaining : Match **List-I** with **List-II**.

Concept : This question is based on no. of atoms.

Solution :

(A) No. of atoms in 1.8 mg water

$$= \frac{1.8 \times 10^{-3}}{18} \times N_A \times 3 = 3 \times 10^{-4} N_A$$

(B) No. of atoms in 9.8 mg H_2SO_4

$$= \frac{9.8 \times 10^{-3}}{98} \times N_A \times 7 = 7 \times 10^{-4} \times N_A$$

(C) No. of atoms in 1.8 mg carbon

$$= \frac{1.8 \times 10^{-3}}{12} \times N_A = 1.5 \times 10^{-4} \times N_A$$

(D) No. of atoms in 5.85 mg NaCl

$$= \frac{5.85 \times 10^{-3}}{58.5} \times N_A \times 2 = 2 \times 10^{-4} N_A$$

Correct match : A-III, B-IV, C-II, D-I

Final Answer : (3)

47. Ans (3)

Explaining : If 0.224 L of H_2 gas is formed at the cathode, the volume of O_2 gas formed at the anode under identical conditions in electrolysis of water is

Concept : This question is based on Faraday's law of electrolysis.

Solution :

$gm\,E_{H_2} = gm\,E_{O_2}$

$0.224 \times 2 = V_L \times 4$

$[V_L = 0.112\ L]$

Final Answer : (3)

48. Ans (4)

Explaining : The plot of lnk vs $\frac{1}{T}$ for the following reaction $2NH_{3\,(g)} \rightarrow N_{2(g)} + 3H_{2(g)}$ gives a straight line with the slope of line equal to -3.0×10^{-5} K. Activation energy for the reaction in J mol^{-1} is -

Concept : This question is based on Arrhenius equation.

Solution :

$k = Ae^{-Ea/RT}$

$\Rightarrow \ln k = \frac{-E_a}{R} \times \frac{1}{T} + \ln A$

$\therefore \frac{-E_a}{R} = -3.0 \times 10^{-5} K$

$\Rightarrow E_a = R \times (3.0 \times 10^{-5} K)$

$\Rightarrow E_a = 8.3\, JK^{-1}mol^{-1} \times 3.0 \times 10^{-5} K$

$\Rightarrow E_a = 2.49 \times 10^{-4}$ J mol^{-1}

Final Answer : (4)

49. Ans (2)

Explaining : 0.05 M ammonium hydroxide solution is dissolved in 0.001 M ammonium chloride solution. What will be the OH^- ion concentration of this solution K_b $(NH_4OH) = 1.8 \times 10^{-5}$

Concept : This question is based on basic buffer solutions.

Solution :

$NH_4OH + NH_4Cl$

0.05 M 0.001 M

$K_b = \frac{[NH_4^+][OH^-]}{[NH_4OH]}$

$[OH^-] = K_b \times \frac{[NH_4OH]}{[NH_4^+]}$

$= 1.8 \times 10^{-5} \times \frac{0.05}{0.001}$

$= 9 \times 10^{-4}$ M

Final Answer : (2)

50. Ans (1)

Explaining : For the reaction

$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$

$K_P = 5 \times 10^{-5}$ at 25°C. $\Delta G°$ for the reaction will be: ($R = 8.31$ J mol^{-1} K^{-1})

Concept : This question is based on relation between ΔG and equilibrium constant.

Solution :

$\Delta G° = -2.303\, RT \log_{10} K_P$

$\Delta G° = -2.303 \times 8.31 \times 298 \times \log(5 \times 10^{-5})$

$\Delta G° = 2.303 \times 8.31 \times 298 \times [5 - \log 5]$

$\Delta G° = 24523$ Joule

$= 24.52$ kJ

Final Answer : (1)

51. Ans (2)

Explaining : When 25% CO_2 reacted the equilibrium pressure developed is 6 atm for the reaction $C(s) + CO_2(g) \rightleftharpoons 2CO(g)$, K_P for the given reaction is :

Concept : This question is based on DOD (α) and K_p.

Solution :

$C(s) + CO_2(g) \rightleftharpoons 2CO(g)$

t_{eq} 0.75P 2×0.25 P

Total pressure $\Rightarrow 0.75\,P + 0.5\,P = 6$

$\Rightarrow P = 4.8$

$K_P = \frac{(0.5\,P)^2}{(0.75\,P)} = 1.6$ atm

Final Answer : (2)

52. Ans (2)

Explaining : In a 0.5 molal solution of KCl, KCl is 50% dissociated. The freezing point of solution will be (K_f =1.86 K kg mol^{-1}) :-

Concept : This question is based on depression in freezing point.

Solution :

$\Delta T_f = i\, m\, K_f$ $i = 1 - \alpha + n\alpha = 1.5$

$= 1.5 \times 0.5 \times 1.86 = 1.395$

$T_f = 273 - 1.395 = 271.60$ K

Final Answer : (2)

53. **Ans (4)**

Explaining : Match the column.

Concept : This question is based on Hydrogen spectrum.

Solution :

(a) First line of Lyman series:

Lyman series: First line: $n_2 = 1 + 1 = 2$

Transition: $n_2 \longrightarrow n_1$. This matches with (S).

(b) Third line of Balmer series:

Balmer series: Third line: $n_2 = 2 + 3 = 5$

Transition: $n_5 \longrightarrow n_2$ .This matches with (R).

(c) Fourth line of Paschen series:

Paschen series: Fourth line: $n_2 = 3 + 4 = 7$

Transition: $n_7 \longrightarrow n_3$ .This matches with (Q).

(d) Second line of Brackett series:

Brackett series: Second line: $n_2 = 4 + 2 = 6$

Transition: $n_6 \longrightarrow n_4$ .This matches with (P).

The correct match is (a) $\longrightarrow$ S, (b) $\longrightarrow$ R, (c) $\longrightarrow$ Q, (d) $\longrightarrow$ P.

Final Answer : (4)

54. **Ans (4)**

Explaining : The equivalent weight of phosphoric acid (H_3PO_4) in the reaction, $NaOH + H_3PO_4 \longrightarrow NaH_2PO_4 + H_2O$ is :-

Concept : This question is based on equivalent weight.

Solution :

Reaction: $NaOH + H_3PO_4 \longrightarrow NaH_2PO_4 + H_2O$

Phosphoric acid (H_3PO_4) is a triprotic acid, but in this reaction, only one proton is neutralized (as indicated by the formation of NaH_2PO_4).

The equivalent weight of H_3PO_4 in this reaction is 98.

Final Answer : (4)

55. **Ans (3)**

Explaining : One mole of an ideal gas at STP expands from 30 L to 40 L in vacuum then work done by the gas will be :-

Concept : This question is based on work done in vacuum.

Solution :

work done: $W = -P_{ext}\Delta V$

in vacuum $P_{ext} = 0$, $W = 0$

Final Answer : (3)

56. **Ans (1)**

Explaining : At 298 K and constant pressure the heat of formation of $C_2H_2(g)$ and $C_6H_6(g)$ is 230 and 85 kJ mol^{-1}. What will be the change in enthalpy for the reaction ?

Concept : This question is based on Heat of formation.

Solution :

$3C_2H_2 \longrightarrow C_6H_6$

$\Delta H = \Delta H_f(C_6H_6) - 3\Delta H_f(C_2H_2)$

$= 85 - 3 \times 230$

$= 85 - 690$

$= -605$ kJ/mol.

Final Answer : (1)

57. **Ans (2)**

Explaining : If four gases A, B, C and D have Henry's constant (K_H) values 1.9×10^1, 6.5×10^{-4}, 7.8×10^{-4} and 6.0×10^3 bar respectively then which of the following gas has maximum solubility at same partial pressure?

Concept : This question is based on Henry's law.

Solution :

Since solubility decreases with increasing K_H, the gas with the lowest K_H has maximum solubility.

Lowest $K_H = 6.5 \times 10^{-4} \longrightarrow$ Gas B

Final Answer : (2)

58. Ans (3)

Explaining : Find ΔG in terms of Faraday ?

Concept : This question is based on Nernst equation.

Solution :

$Zn + Cu^{+2} \longrightarrow Zn^{+2} + Cu$; $E_{Cell} = 1.1 - \frac{0.06}{2} = 1.07$ V

$\Delta G = -2F(1.07) = -2.14$ F

Final Answer : (3)

59. Ans (4)

Explaining : The rate constant is numerically the same for three reactions of first, second and third order respectively. Which of the following is correct?

Concept : This question is based on Rate law.

Solution :

Rate constant same for

I order Reaction

II order Reaction

III order Reaction

Case I [A] = 1

$r = k[A]^1$ for I order reaction

$r_1 = k[1]$

$r_2 = k[A]^2$ for II order reaction

$r_2 = k[1]^2$

$r_3 = k[1]^3$ for III order reaction

$\boxed{r_1 = r_2 = r_3}$

Case II [A] < 1

Let [A] = 0.9

$r_1 = k[0.9]$ for I order reaction

$r_2 = k[0.9]^2$ for II Order reaction

$r_3 = k[0.9]^3$ for III order reaction

$\boxed{r_3 < r_2 < r_1}$

Case-III

[A] > 1 Let [A] = 1.1

$r_1 = k[1.1]$ for I order

$r_2 = k[1.1]^2$

$r_2 = k[1.21]$ for II order

$r_3 = k[1.1]^3$

$r_3 = k[1.331]$ for III order

$\boxed{r_3 > r_2 > r_1}$

Final Answer : (4)

60. Ans (2)

Explaining : The energy of an electron in an excited H-atom is -1.51 eV. Angular momentum of electron in the given orbit will be

Concept : This question is based on Bohr's Atomic model.

Solution :

$E = -13.6 \times \frac{Z^2}{n^2}$, Here $E = -1.51$ eV, $Z = 1$

$n = 3$

Angular momentum $= \frac{nh}{2\pi} = \frac{3h}{2\pi}$

Final Answer : (2)

61. Ans (2)

Explaining : The transition metal having highest 3rd ionisation enthalpy is :

Concept : This question is based on ionisation enthalpy.

Solution :

3rd Ionisation energy : [NCERT Data]

V : 2833 KJ/mol

Cr : 2990 KJ/mol

Mn : 3260 KJ/mol

Fe : 2962 KJ/mol

alternative

Mn : $3d^5\ 4s^2$

Fe : $3d^6\ 4s^2$

Cr : $3d^5\ 4s^1$

V : $3d^3\ 4s^2$

So Mn has highest 3rd IE among all the given elements due to d^5 configuration.

Final Answer : (2)

62. Ans (1)

Explaining : A/R given for nature of oxide.

Concept : This question is based on Nature of oxide.

Solution :

Cr^{2+} is reducing as it configuration changes from d^4 to d^3 due to formation of Cr^{3+}, which has half filled t_{2g} level, on other hand, the change Mn^{3+} to Mn^{2+} result half filled d^5 configuration which has extra stability.

Final Answer : (1)

63. **Ans (2)**

Explaining : Statement given for Metals & Non metals.

Concept : This question is based on Nature of elements.

Solution :

I. In p-Block both metals and non metals are present but in d-Block only metals are present.

II. EN and IE of non metals are greater than that of metals

I-False, **II**-True

Final Answer : (2)

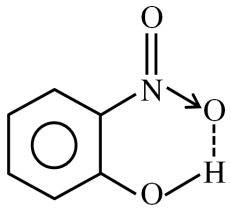
64. **Ans (4)**

Explaining : Select the compound from the following that will show intramolecular hydrogen bonding.

Concept : This question is based on H-Bonding.

Solution :

H_2O, NH_3, C_2H_5OH ⇒ Intermolecular H-Bonding



⇒ Intramolecular H-Bonding

Final Answer : (4)

65. **Ans (3)**

Explaining : Statement given for nature of oxide & catenation property.

Concept : This question is based on Nature of oxide.

Solution :

SiO_2 and GeO_2 are acidic and SnO, PbO are amphoteric.

Carbon does not have d-orbitals so can not form pπ-dπ Bond with itself. Due to properties of catenation and pπ-pπ bond formation carbon is able to show allotropic forms.

Final Answer : (3)

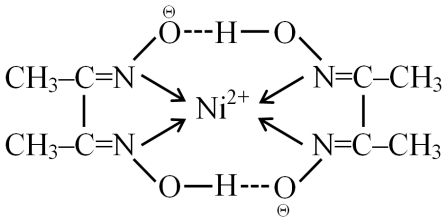
66. **Ans (1)**

Explaining : Statement given for complex compound.

Concept : This question is based on chelating & O.N.

Solution :

Ni^{2+} + NH_4OH + dmg ⟶



2 Five member ring

III II

$Fe_4[Fe(CN)_6]_3$

Prussian Blue

Final Answer : (1)

67. **Ans (2)**

Explaining : Which order of dipole moment is correct ?

Concept : This question is based on dipole moment.

Solution :

Correct order of dipole moment :

$CCl_4 < CHCl_3 < CH_2Cl_2 < CH_3Cl$

$BF_3 < NF_3 < NH_3$

O-dichlorobenzene > m-dichlorobenzene > p-dichlorobenzene

Final Answer : (2)

68. **Ans (3)**

Explaining : Which of the following is correct on the basis of MOT ?

Concept : This question is based on MOT.

Solution :

	O_2	NO	N_2
unpaired ⇒ e^-	2	1	0

Final Answer : (3)

69. Ans (4)

Explanation the Question : Identify the only correct chemical statement from the four options.

Concept: This question is based on electron Affinity.

Solution: Due to its small size and high inter-electronic repulsion in the compact 2p orbitals, Fluorine's electron affinity is anomalously lower than that of Chlorine.

Final Answer: (4)

70. Ans (2)

Explaining : Which of the following is not correctly matched :-

Concept : This question is based on Isomerism in complex compound & Magnetic nature.

Solution :

$[Pt(NO_2)(NH_3)(H_2O)Cl]$

$\Downarrow$

dsp^2 (Square planar)

Not show O.I.

Final Answer : (2)

71. Ans (2)

Explaining : Match the column-

Concept : This question is based on Periodic properties.

Solution :

$$\text{Size} \propto \frac{(-)\text{ ve charge}}{(+)\text{ ve charge}} \propto \frac{1}{\text{I.E.}} \propto \frac{1}{\text{E.N.}} \propto \frac{1}{Z_{eff}}$$

Final Answer : (2)

72. Ans (3)

Explaining : Which of the following order is not correct ?

Concept : This question is based on acidic nature.

Solution :

The order of Lewis acidity ($BF_3 < BCl_3 < BBr_3$), Lewis basicity ($O(CH_3)_2 > O(SiH_3)_2$) orders are correct. However, acidic strength ($H_3PO_4 > H_3PO_3 > H_3PO_2$) is **incorrect**.

Final Answer : (3)

73. Ans (3)

Explaining : Which of the following has banana bond :

Concept : This question is based on banana bond.

Solution :

Banana bond $\Rightarrow 3C - 2e^-$

$Al_2(CH_3)_6$

$Al_2Cl_6 \Rightarrow 3C - 4e^-$

$C_2(CH_3)_6$ & $Si_2(CH_3)_6$

$\Downarrow$

$2C - 2e^-$

Final Answer : (3)

74. Ans (3)

Explaining : Select the correct statement :-

Concept : This question is based on oxidizing power & stability (d-block)

Solution :

(a) $Mn^{+3} \longrightarrow Mn^{+2}$ $E° = +1.51$ V (Highly positive)

(b) MnF_7, BF_6^{-3} (does not exist) but SiF_6^{2-} exists.

(c) $La^{+3}(Xe)$ $Lu^{+3}(Xe)4f^{14}$

(d) CrO_5 – Deep blue

Final Answer : (3)

75. Ans (3)

Explaining : Which of the following is correct :-

Concept : This question is based on Alloy.

Solution :

German silver $\longrightarrow$ Cu + Zn + Ni

Brass – Cu + Zn

Gunmetal $\longrightarrow$ Cu + Zn + Sn

Final Answer : (3)

76. Ans (3)

Explaining : S_N2 reaction

Concept : This question is based on S_N2 reaction

Solution :

$CH_3CH_2CH_2CH_2Cl$ > $(CH_3)_2CHCH_2Cl$ > $CH_3CH_2CH(Cl)CH_3$ > $(CH_3)_3CCl$

$$\text{Rate of } S_N2 \propto \frac{1}{\text{Steric Hindrance}}$$

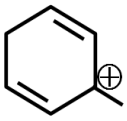
Final Answer : (3)

77. **Ans (4)**

Explaining : Which of the following is the most stable carbocation.

Concept : This question is based on Most stable carbocation.

Solution :



3° Cation + resonance stabilized.

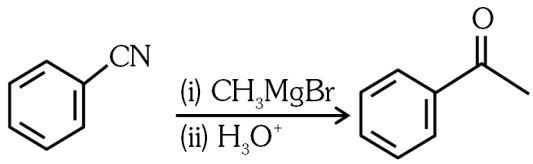
Final Answer : (4)

78. **Ans (4)**

Explaining : The incorrect method to synthesize benzaldehyde is.

Concept : This question is based on Benzaldehyde synthesize

Solution :



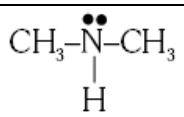
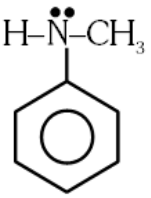
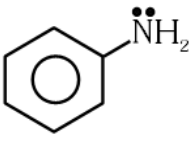
Final Answer : (4)

79. **Ans (1)**

Explaining : Match List-I with List-II.

Concept : This question is based on Match List-I with List-II.

Solution :

	Amines	pK_b
(a)		3.27
(b)		4.75
(c)		9.30
(d)		9.38

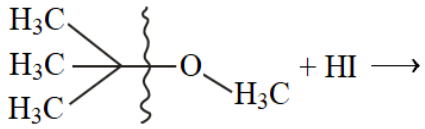
Final Answer : (1)

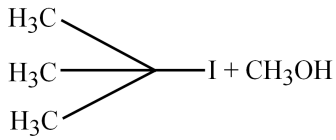
80. **Ans (2)**

Explaining : The major products formed in the following reaction are.

Concept : This question is based on Ether cleavage reaction.

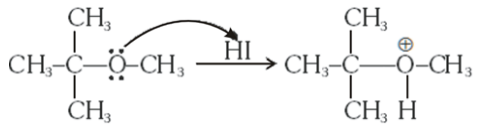
Solution :

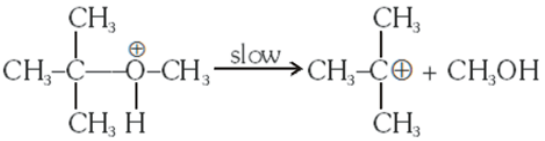




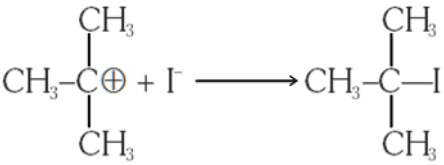
Whenever one of the alkyl group is a tertiary group, the halide formed is a tertiary halide.

Mechanism :

(1) 

(2) 

(stable 3 carbocation)

(3) 

2-Iodo-2-Methylpropane

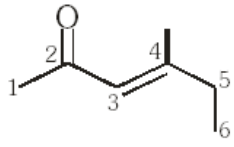
Final Answer : (2)

81. **Ans (1)**

Explaining : The correct IUPAC name of the following compound is.

Concept : This question is based on IUPAC

Solution :



Longest parent chain is selected by giving lowest numbering to ketone **IUPAC name** – 4-methyl hex-3-en-2-one

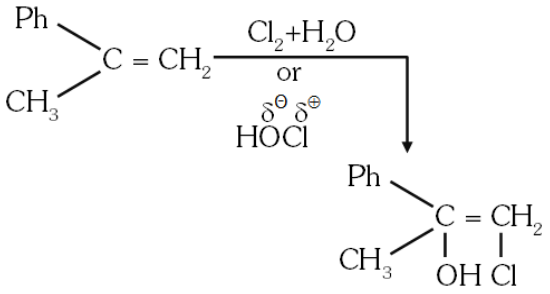
Final Answer : (1)

82. **Ans (1)**

Explaining : Compound (A) is

Concept : This question is based on EAR

Solution :



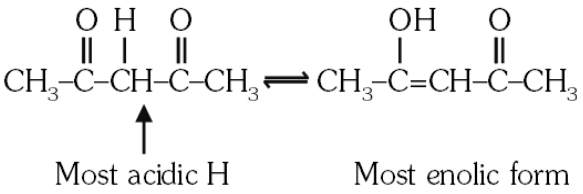
Final Answer : (1)

83. **Ans (2)**

Explaining : The most stable enolic form of 2,4-pentanedione is.

Concept : This question is based on Most stable enolic form

Solution :



Final Answer : (2)

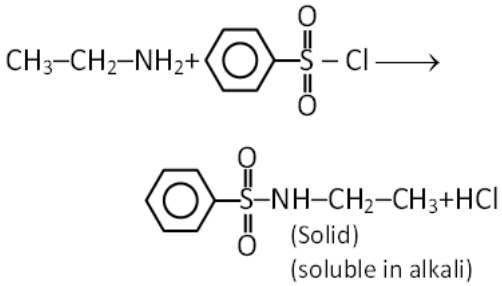
84. **Ans (3)**

Explaining : dentify the compound that will react with Hinsberg's reagent to give a solid which dissolves in alkali.

Concept : This question is based on Hinsberg's reagent

Solution :

1° amines react with Hingsberg's reagent to give a solid, which dissolve in alkali.



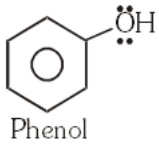
Final Answer : (3)

85. **Ans (4)**

Explaining : Which of the following compound is most reactive in electrophilic aromatic substitution.

Concept : This question is based on Most reactive in Electrophilic aromatic substitution.

Solution :



+ R effect of –OH group enhances the reactivity in aromatic electrophilic substitution reaction.

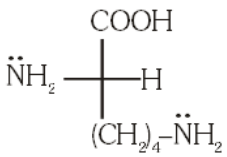
Final Answer : (4)

86. **Ans (1)**

Explaining : Which of the following is a basic amino acid

Concept : This question is based on Basic Amino acid

Solution :



Since it contains more number of $-NH_2$ groups as compared to –COOH groups hence it is basic amino acid.

Final Answer : (1)

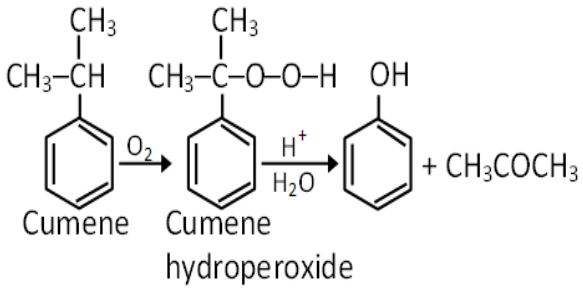
87. **Ans (2)**

Explaining : The structure of intermediate A in the following reaction is.

Concept : This question is based on Find Intermediate in reaction

Solution :

Phenol is manufactured from the hydrocarbon, cumene. Cumene (isopropylbenzene) is oxidised in the presence of air to cumene hydroperoxide. it is converted to phenol and acetone by treating it with dilute acid. Acetone, a by-product of this reaction, is also obtained in large quantities by this method.



Final Answer : (2)

88. **Ans (2)**

Explaining : Assertion and reason based

Concept : This question is based on Inductive effect

Solution :

For Halogen : –I > + M ⟶ ortho, para directing

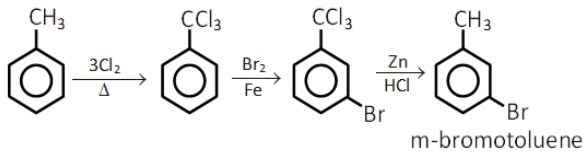
Final Answer : (2)

89. **Ans (1)**

Explaining : The compound C_7H_8 undergoes the following reactions, The product 'C' is.

Concept : This question is based on Find A, B, C

Solution :



Final Answer : (1)

90. **Ans (3)**

Explaining : The most suitable method of separation of 1 : 1 mixture of ortho and para-nitrophenols is.

Concept : This question is based on Steam distillation.

Solution :

The ortho and para isomers can be separated by steam distillation o-Nitrophenol is steam volatile due to intramolecular hydrogen bonding while p-nitro phenol is less volatile due to intermolecular hydrogen bonding which cause association of molecule.

Final Answer : (3)

91. **Ans (2)**

Explanation: This question is based on the pollination mechanism of the Ophrys orchid, which uses sexual deceit for pollination.

- One petal of the flower resembles the female bee/bumble bee in shape, colour and markings.
- The male bee gets attracted and attempts pseudocopulation with the flower.
- During this process, pollen grains are transferred, leading to pollination.

Therefore:

- Statement (1) ⟶ Correct
- Statement (2) ⟶ Incorrect, because pollination is carried out by the male bee, not the female.
- Statement (3) ⟶ Correct
- Statement (4) ⟶ Correct

Final Answer : (2)

92. Ans (4)

Explanation: Assertion is false because mycorrhiza is a mutualistic association between fungi and plant roots.

The fungus gets food from the plant.

The plant receives water and minerals from the fungus.

Reason is also false because in lichen, both partners (alga and fungus) benefit from each other.

The alga prepares food by photosynthesis.

The fungus provides shelter, water and minerals.

Thus, both the Assertion and the Reason are false.

Final Answer : (4)

93. Ans (3)

Solution : Humification is the process in which decomposed organic matter is converted into a dark-coloured amorphous substance called humus. Humus is rich in nutrients and highly resistant to microbial action.

Other options:

Fragmentation ⟶ Breakdown of detritus into smaller particles by detritivores.

Leaching ⟶ Water-soluble nutrients move deeper into the soil.

Mineralization ⟶ Release of inorganic nutrients from humus.

Final Answer : (3)

94. Ans (4)

Explanation: The **generative cell** is spindle-shaped and floats freely in the cytoplasm of the vegetative cell within the pollen grain. It later divides to form two male gametes.

Final Answer : (4)

95. Ans (3)

Explanation: In **Drosophila**, the **XX chromosome condition** produces a female fly, while the XY condition produces a male fly.

Other options are incorrect because:

- Grasshopper males are **XO**, not XX.
- Klinefelter's syndrome in humans is **XXY**, not XO.

In birds, **ZZ** determines male sex and **ZW** determines female sex.

Final Answer : (3)

96. Ans (3)

Explanation: Turner's syndrome occurs due to **monosomy of the X chromosome (XO condition)**, where an individual possesses only one X chromosome.

Karyotype: **45, XO**

Final Answer : (3)

97. Ans (2)

Explanation: Genes showing **50% recombination** behave as if they are independently assorting.

This occurs when:

- Genes are located on **non-homologous chromosomes**, or
- Genes are very far apart on the same chromosome.

Therefore, such genes are **not tightly linked**. Hence, the statement "The genes are very tightly linked" is incorrect.

Final Answer : (2)

98. Ans (2)

Explanation: Ethylene is responsible for the formation of the **apical hook** in dicot seedlings. The apical hook protects the delicate shoot tip while the seedling emerges through the soil.

Final Answer : (2)

99. **Ans (1)**

Explanation: Sugarcane is a **C_4 plant**.

In C_4 plants:

- **Mesophyll cells** contain **PEPCase** and initial fixation forms **OAA (Oxaloacetic acid)**.
- **Bundle sheath cells** contain **RuBisCO**, where the **C_3 cycle (Calvin cycle)** occurs.

Therefore, Option (1) correctly matches the processes occurring in mesophyll and bundle sheath cells.

Final Answer : (1)

100. **Ans (1)**

Explanation: During non-cyclic photophosphorylation:

- Electrons move from **PSII to ETS** in a **downhill** manner.
- Absorption of light in **PSI** raises the electron energy level again, which is an **uphill** movement.
- Finally, electrons reduce $NADP^+$ to NADPH in a **downhill** manner.

Thus, the sequence is:**Uphill ⟶ Downhill ⟶ Uphill ⟶ Downhill**

Final Answer : (1)

101. **Ans (1)**

Explanation: In grasses (monocots), the guard cells of stomata are **dumb-bell shaped**.This shape helps in efficient opening and closing of stomata.

Final Answer : (1)

102. **Ans (4)**

Explanation: In the fluid mosaic model of the plasma membrane:

- **Integral proteins** are embedded within the lipid bilayer.
- **Cholesterol** is present between phospholipid molecules and provides membrane stability.
- **Sugar molecules** are attached externally as glycoproteins or glycolipids.

According to the given diagram:

- Integral protein ⟶ **S**
- Cholesterol ⟶ **P**
- Sugar ⟶ **R**

Final Answer : (4)

103. **Ans (4)**

Explanation: After **Telophase I of meiosis**, homologous chromosomes separate, reducing the chromosome number to half.

Therefore:

- Each daughter cell becomes **haploid (n)**.
- Each chromosome still consists of **two sister chromatids**, because sister chromatids separate only during Meiosis II.

Final Answer : (4)

104. **Ans (1)**

Explanation: Top carnivores such as lions and tigers occupy the highest trophic level and receive the least amount of energy.

According to the **10% law of energy transfer**, only about 10% of energy passes from one trophic level to the next. As a result, very little energy reaches top carnivores, making their population smaller and more vulnerable.

Thus:

- Assertion is true.
- Reason is true.
- Reason correctly explains the Assertion.

Final Answer : (1)

105. **Ans (1)**

Explanation: **Ex situ conservation** involves conservation of organisms outside their natural habitat, such as:

- Botanical gardens
- Wildlife safari parks
- Seed banks

Therefore:

- **B, D, E ⟶ Ex situ conservation In situ conservation**

involves protection of organisms within their natural habitat, such as:

- National parks
- Sacred groves

Therefore:

- **A, C ⟶ In situ conservation**

Final Answer : (1)

106. **Ans (3)**

Explanation: Statement (b) is correct because sunflower belongs to the family Compositae (Asteraceae) and is commonly grown as an ornamental plant.
Statement **(d)** is correct because the given floral formula belongs to the **Malvaceae** family.
Statements **(a)** and **(c)** are incorrect because:

- In **Liliaceae**, the androecium is **epitepalous**, not epipetalous.
- In **Solanaceae**, the androecium is **epipetalous**, not epitepalous.

Final Answer : (3)

107. **Ans (3)**

Explanation: Dark reactions (enzymatic reactions of the Calvin cycle) are more sensitive to temperature changes than light reactions.
Light reactions mainly depend on light and are comparatively less affected by temperature.Therefore, the statement “Light reactions are more sensitive to temperature than dark reactions” is incorrect.
Final Answer : (3)

108. **Ans (3)**

Explanation: The developmental sequence in a plant cell is:

1. **Cell division** – increase in cell number through mitosis.
2. **Plasmatic growth** – increase in protoplasm and cell size.
3. **Maturation** – cells attain their final shape and function.
4. **Senescence** – ageing phase of the cell.

Thus:

- A ⟶ Cell division
- B ⟶ Plasmatic growth
- C ⟶ Maturation
- D ⟶ Senescence

Final Answer : (3)

109. **Ans (4)**

Explain Question: The Question asks to select the correct statements about fungi.

Concept: Fungi

Solution: The correct statement includes:

(A) Most of the fungi are saprophytes.

(B) Fungi can also survive as symbionts.

(C) Sexual reproduction in fungi involves three steps.

(D) Reduction division in fungi occurs in fruiting bodies which form haploid spores.

So, All statements A, B, C & D are correct and the correct answer is: 4.

Final answer: 4.

110. **Ans (2)**

Explanation: Adaptations promoting **self-pollination** include:

- **Monocliny (bisexuality)** → Both male and female reproductive organs are present in the same flower.
- **Cleistogamy** → Flowers remain closed, ensuring self-pollination.

Whereas:

- **Chasmogamy** promotes cross-pollination because flowers open normally.
- **Protandry** prevents self-pollination by different maturation times of anthers and stigma.

Final Answer : (2)

111. **Ans (2)**

Explanation: The Assertion is true because in populations consisting of organisms of greatly different sizes (e.g., Parthenium plants and a large banyan tree), **biomass** gives a more meaningful estimate than numerical density.

The Reason is also true because tiger census is estimated using **pug marks and fecal pellets**.

However, the Reason does not explain the Assertion.

Final Answer : (2)

112. **Ans (2)**

Solution : In any ecosystem, the **energy pyramid is always upright** because energy decreases progressively at each trophic level according to the 10% law.

Even if the number pyramid may vary, the energy available to higher trophic levels is always less than that of lower trophic levels.

Final Answer : (2)

113. **Ans (1)**

Explanation: Humus decomposes at an extremely slow rate because it is highly resistant to microbial action.

Therefore:

- Assertion is true.
- Reason is true.
- Reason correctly explains the Assertion.

Final Answer : (1)

114. **Ans (4)**

Explanation: The pie chart represents the relative diversity of vertebrates.

Given:

- A = Birds
- D = Fishes

The remaining categories correspond to:

- C = Amphibians
- E = Mammals

Final Answer : (4)

115. **Ans (3)**

Explanation: The five-carbon intermediate compound of the TCA (Krebs) cycle is **α-ketoglutaric acid**.

Carbon count of given compounds:

- Citric acid → 6C
- Succinyl CoA → 4C
- α-ketoglutaric acid → 5C
- Fumaric acid → 4C

Final Answer : (3)

116. **Ans (1)**

Explanation: Kranz anatomy is a characteristic feature of **C_4 plants**.

- In C_4 plants, large bundle sheath cells surround the vascular bundles, often forming several layers.

Therefore:

- Statement-I is correct.
- Statement-II is also correct.

Final Answer : (1)

117. **Ans (2)**

Explanation: A–iv, B–ii, C–i, D–iii, E–vi, F–v

- ⊕ represents **actinomorphic symmetry** ⟶ iv
- % represents **zygomorphic symmetry** ⟶ ii
- G̲ represents **inferior ovary** ⟶ i
- G represents **superior ovary** ⟶ iii
- Epipetalous stamens ⟶ v
- Epitepalous stamens ⟶ vi

Thus, the correct matching is:

Final Answer : (2)

118. **Ans (4)**

Explanation: A pentapeptide consists of **5 amino acids** joined by **4 peptide bonds**.

Given:

- Molecular weight of one amino acid = 180

Total molecular weight before bond formation:for tetrapeptide 4 amino acids 4 x 180 = 720

Each peptide bond releases one water molecule (18 u):

3times 18 = 54 Final molecular weight:

720 - 54 = 666

Final Answer : (4)

119. **Ans (3)**

Explanation: Tryptophan is an aromatic amino acid because it contains an aromatic indole ring in its side chain.

Other options:

- Glutamic acid ⟶ acidic amino acid
- Lysine ⟶ basic amino acid
- Glycine ⟶ simplest non-aromatic amino acid

Final Answer : (3)

120. **Ans (2)**

Explanation: During the **zygotene stage of Prophase-I**, homologous chromosomes pair to form **bivalents**.

Given:

- Total chromosomes in G1 phase = 20 (2n = 20)

Therefore: [n = 10] Hence, the number of bivalents formed in zygotene = **10**.

Final Answer : (2)

121. **Ans (3)**

Explanation: In humans:

- Total chromosomes = 46
- Autosomes = 44
- Sex chromosomes in males = XY

Therefore, the chromosomal constitution of a human male is: [44 Autosomes + XY]

Final Answer : (3)

122. **Ans (2)**

Explanation: The X chromosome participates in sex determination; therefore, it is called a **sex chromosome** (allosome). Autosomes are chromosomes not involved in sex determination.

Final Answer : (2)

123. **Ans (3)**

Explanation: According to Chargaff's rule in double-stranded DNA:

- %C = %G
- %A = %T

Given: [C = 12%]

Therefore: [G = 12%]

Remaining percentage: [100 - (12 + 12) = 76%]

Thus: [A = T = 38%]

Final Answer : (3)

124. **Ans (2)**

Explanation:

- (a) Correct ⟶ A genus is a group of closely related species sharing more similarities with one another than with species of other genera.
- (b) Correct ⟶ Classification involves multiple hierarchical steps.
- (c) Incorrect ⟶ *Solanum*, *Petunia* and *Datura* belong to the family **Solanaceae**, not Convolvulaceae.
- (d) Correct ⟶ Taxonomic categories are distinct biological entities as well as morphological groups.

Final Answer : (2)

125. **Ans (1)**

Explanation: Gymnosperms are **heterosporous**, producing two types of spores:

- Microspores (male)
- Megaspores (female)

Therefore, the statement "They are homosporous" is incorrect.Other statements are correct because:

- Male and female gametophytes are dependent on the sporophyte.
- Most gymnosperms possess needle-like leaves with thick cuticle.
- Seeds are naked (not enclosed within fruits).

Final Answer : (1)

126. **Ans (2)**

Explanation: Tobacco Mosaic Virus (TMV) possesses:

- A **coiled single-stranded RNA**
- Protein subunits called **capsomeres** arranged helically around the RNA.

Therefore, it shows both a coiled RNA strand and capsomeres.

Final Answer : (2)

127. **Ans (1)**

Explanation: In a typical mature embryo sac:

- Two **synergids** and one **egg cell** form the egg apparatus at the micropylar end.
- The **central cell** contains two polar nuclei.
- Three **antipodal cells** are present at the chalazal end.

According to the diagram:

- A ⟶ Synergids
- B ⟶ Egg
- C ⟶ Central cell
- D ⟶ Antipodals

Final Answer : (1)

128. **Ans (1)**

Explanation: The Western Ghats–Sri Lanka region, Indo-Burma region and the Himalayas are recognized as biodiversity hotspots because they possess:

- High species richness
- High degree of endemism

Therefore:

- Assertion is true.
- Reason is true.
- Reason correctly explains the Assertion.

Final Answer : (1)

129. **Ans (4)**

Explanation: Statement (I) is incorrect because O_2, not CO_2, helps in releasing metabolic energy during respiration.

Statement (II) is correct because nutrients are required for protoplasm synthesis and energy-related activities.

Statement (III) is incorrect because optimum, not necessarily low, temperature is required for plant growth.

Statement (IV) is correct because environmental signals such as light and gravity influence growth.

Final Answer : (4)

130. **Ans (3)**

Explanation: In one turn of the Krebs cycle, **2 molecules of CO_2** are released.

Since one glucose molecule produces **2 acetyl CoA molecules**, the Krebs cycle runs twice per glucose molecule.

Final Answer : (3)

131. **Ans (1)**

Explanation: Heterophylly is the occurrence of different types of leaves on the same plant and is an example of **plasticity**.

In Buttercup (*Ranunculus*), heterophylly occurs due to environmental conditions:

- Submerged leaves are finely dissected.
- Aerial leaves are broader.

Thus:

- Statement-I is correct.
- Statement-II is also correct.

Final Answer : (1)

132. **Ans (2)**

Explanation: In roots, the primary xylem arrangement is **exarch**, which means:

- Protoxylem lies towards the periphery.
- Metaxylem lies towards the centre.

The Assertion is correct.

However, the Reason is incorrect because it states the reverse arrangement.

Final Answer : (2)

133. **Ans (3)**

Explanation: The given structure represents **cholesterol**, which is a sterol component of animal cell membranes.Characteristic features include:

- Four fused hydrocarbon rings
- A hydroxyl (–OH) group
- A hydrocarbon tail

Final Answer : (3)

134. **Ans (4)**

Explanation: Single membrane-bound organelles include:

- Lysosomes
- Golgi apparatus
- Endoplasmic reticulum

Whereas:

- Nucleus and mitochondria are double membrane-bound.
- Ribosomes are non-membranous.

Final Answer : (4)

135. **Ans (2)**

One pollen mother cell undergoes one meiotic division to produce **4 pollen grains**.

To produce 500 pollen grains:

Therefore, **125 meiotic divisions** are required.

Final Answer : (2)

136. **Ans (1)**

Explanation: DNA maintains a uniform diameter because:

- A **purine** (A or G) always pairs with a **pyrimidine** (T or C).

Thus:

- A–T pairing and G–C pairing maintain equal width throughout the DNA molecule.

Therefore:

- Assertion is true.
- Reason is true.
- Reason correctly explains the Assertion.

Final Answer : (1)

137. **Ans (2)**

Solution : (a)–(iii), (b)–(iv), (c) - (ii), (d) - (i)

	List - I		**List - II**
(a)	*Chara*	(iii)	Algae
(b)	*Pinus*	(iv)	Gymnosperm
(c)	*Selaginella*	(ii)	Lycopsida
(d)	*Sphagnum*	(i)	Moss

Final Answer : (2)

138. **Ans (4)**

Question explanation: Questions asking about members of phycomycetes are found in which places,select the correct answer according to the given option.

Concept: Class-Phycomycetes

Solution: Members of phycomycetes are found in aquatic habitats and on decaying wood in moist and damp places or as obligate parasites on plants.

Final answer : (4)

139. **Ans (3)**

Explain Question : The question is asking you to match each cell structure (List I) with its correct description or location (List II).

Concept: Cell structure

Solution : (a) Cristae ⟶ infoldings in mitochondria ⟶ (iii)

(b) Thylakoids ⟶ flattened membranous sacs in stroma of plastids (chloroplasts) ⟶ (iv)

(c) Centromere ⟶ primary constriction in chromosome ⟶ (i)

(d) Cisternae ⟶ disc-shaped sacs in Golgi apparatus ⟶ (ii)

The Correct Answer: Option 3 — (a)–(iii), (b)–(iv), (c)–(i), (d)–(ii)

Final answer : (3)

140. **Ans (1)**

Question Explanation:

Assertion (A) : Meiosis ensures the production of haploid phase in the life cycle of sexually reproducing organism.

Reason (R) : Meiosis involve reduction in chromosome number to half as that of parent cell.

Solution:

Assertion (A) is true: Meiosis ensures the production of haploid cells, which is essential for maintaining the correct chromosome number across generations in sexually reproducing organisms. The fusion of haploid gametes during fertilization restores the diploid chromosome number in the offspring.

Reason (R) is also true: Meiosis involves two rounds of cell division but only one round of DNA replication, resulting in daughter cells with half the chromosome number of the parent cell.

The Reason accurately explains why meiosis produces a haploid phase in the life cycle by reducing the chromosome number to half, ensuring genetic stability across generations.

Therefore, the correct answer is that both Assertion and Reason are true, and the Reason is a correct explanation of the Assertion.

Hence,

Correct option: (1) Both Assertion and Reason are True and the Reason is a correct explanation of the Assertion.

141. **Ans (4)**

Explanation: In Metaphase-I of meiosis, homologous chromosomes pair to form **bivalents (chromosome pairs)** at the equatorial plate.

Given:

[n = 25] Therefore: Number of homologous chromosome pairs (bivalents) = **25**

Final Answer : (4)

142. **Ans (4)**
Solution : Both DNA and RNA are able to mutate, but DNA mutates at a much faster rate.
Final Answer : (4)

143. **Ans (4)**
Solution : Both Assertion and Reason are false.
Final Answer : (4)

144. **Ans (2)**
Solution : Statement I is correct but Statement II is incorrect
Final Answer : (2)

145. **Ans (1)**
Solution :

	Column-I		Column-II
(a)	Placenta	(ii)	Human Chorionic Gonadotropin(hCG)
(b)	Zona pellucida	(iii)	Layer of the ovum
(c)	Bulbo-urethral glands	(iv)	Lubrication of the Penis
(d)	Leydig's cells	(i)	Androgens

Final Answer : (1)

146. **Ans (1)**
Solution : anterior pituitary gland and stimulates secretion of LH and FSH.
Final Answer : (1)

147. **Ans (3)**
Solution : Either zygote or early embryo upto 8 cell stage
Final Answer : (3)

148. **Ans (3)**
Solution : Forelimbs of cat, lizard used in walking; forelimbs of whale used in swimming and forelimbs of bats used in flying are an example of homologous organs
Final Answer : (3)

149. **Ans (2)**
Solution : (a), (b) and (c) only
Final Answer : (2)

150. **Ans (3)**
Solution : Gene therapy was given in 1990 to a four year old girl with adenosine deaminase (ADA) deficiency
Final Answer : (3)

151. **Ans (2)**
Solution : amp^R, tet^R-antibiotic resistance genes
Final Answer : (2)

152. **Ans (4)**
Solution : IgA immunoglobulins does constitute the largest percentage in human milk
Final Answer : (4)

153. **Ans (2)**
Solution :

Column I	*Column II*
(a) Amoebiasis	(ii) Use only sterilized food and water
(b) Diphtheria	(iii) DPT Vaccine
(c) Cholera	(iv) Use oral rehydration therapy
(d) Syphilis	(i) *Treponema pallidum*

Final Answer : (2)

154. **Ans (2)**
Solution : Sea horse and Flying fish — Cold blooded (poikilothermal): Both animals belong to the class Osteichthyes (bony fishes). All fishes are ectothermic (cold-blooded), meaning they cannot regulate their internal body temperature and rely on environmental temperatures
Final Answer : (2)

155. **Ans (3)**
Solution : B -Glandular epithelium - Intestine - Secretion
Final Answer : (3)

156. **Ans (4)**

Solution : SA node has the highest rate of depolarisation.

Final Answer : (4)

157. **Ans (4)**

Solution : D - Posterior pituitary - Release oxytocin and vasopressin

Final Answer : (4)

158. **Ans (2)**

Solution : Each of the fore limbs and hind limbs end in five digits

Final Answer : (2)

159. **Ans (2)**

Solution : The statement "External ear and tympanum can be seen externally" is false

In the biological context of animal anatomy (most notably regarding amphibians like frogs), external ears are absent, and only the tympanum (eardrum) can be seen externally on the surface of the skin

Final Answer : (2)

160. **Ans (4)**

Solution : Vasectomy - prevents spermatogenesis

Final Answer : (4)

161. **Ans (2)**

Solution : Transgenic 'Cow Rosie' was nutritionally more balanced product for human babies than natural cow milk because it contained human alpha–lactalbumin

Final Answer : (2)

162. **Ans (4)**

Solution : Post anal part, Notochord, Nerve cord and Gill slits

Final Answer : (4)

163. **Ans (2)**

Solution :

	Column I		**Column II**
A	Inspiratory capacity (IC)	1	Total air, a person can inspire after normal expiration
B	Expiratory capacity (EC)	4	Maximum volume of air a person can expire after a normal inspiration
C	Functional residual capacity (FRC)	3	Volume of the air that will remain in lungs after a normal expiration
D	Vital capacity (VC)	2	Maximal volume of the air, a person can breath in after a forced expiration

Final Answer : (2)

164. **Ans (3)**

Solution : In reality, impulse transmission across an **electrical synapse is always faster** than that across a chemical synapse.

Final Answer : (3)

165. **Ans (2)**

Solution : Assertion (A) is true, but Reason (R) is false.

Final Answer : (2)

166. **Ans (4)**

Solution : *Trichoderma* species against several plant pathogens.

Final Answer : (4)

167. **Ans (3)**

Solution : A-*Lactobacillus;* B-*Trichoderma;* C-*Penicillium notatum;* D-*Monascus*

Final Answer : (3)

168. **Ans (1)**

Solution :

List-I		List-II	
(a)	Diaphragms	(iv)	They cover the cervix blocking the entry of sperms
(b)	Contraceptive Pills	(i)	Inhibit ovulation and Implantation
(c)	Intra uterine Devices	(ii)	Increase phagocytosis of sperm within Uterus
(d)	Lactational Amenorrhea	(iii)	Absence of Menstrual cycle and ovulation

Final Answer : (1)

169. **Ans (3)**

Solution : Polymorpho-nuclear leukocytes and monocytes - Cellular barriers

Final Answer : (3)

170. **Ans (3)**

Solution : Variation in gene frequencies within populations can occur by chance rather than by natural selection. This is referred to as genetic drift

Final Answer : (3)

171. **Ans (3)**

Solution : Flippers of penguins and dolphins are a classic example of analogous organs, not homologous organs.

Final Answer : (3)

172. **Ans (1)**

Solution : C-Extension in the presence of heat stable DNA polymerase

Final Answer : (1)

173. **Ans (2)**

Solution :

(a) Restriction endonuclease	(iii) Cuts DNA at specific position
(b) Restriction exonuclease	(iv) Removes nucleotides from the ends of DNA
(c) DNA ligase	(i) Joins the DNA fragements
(d) Taq polymerase	(ii) Extends primers on genomic DNA template

Final Answer : (2)

174. **Ans (1)**

Solution : Statements (A) and (B)

Final Answer : (1)

175. **Ans (3)**

Solution :

(a)	*Bacillus thuringiensis*	(iv)	Cry proteins
(b)	*Thermus aquaticus*	(iii)	DNA polymerase
(c)	*Agrobacterium tumefaciens*	(i)	Cloning vector
(d)	*Salmonella typhimurium*	(ii)	Construction of first rDNA molecule

Final Answer : (3)

176. **Ans (1)**

Solution : Mouth ventral, gills without operculum; skin with placoid scales; persistent notochord - Chondrichthyes

Final Answer : (1)

177. **Ans (3)**

Question Explanation: The question asks which statements are correct for amphibians.

Concept: Amphibia

Solution: The correct answer is option 3. Statements B, C, and D are correct. Amphibian skin is moist without scales, their body is divisible into head and trunk, and they have a three-chambered heart.

Final Answer: 3

178. Ans (1)

Solution:

	'A'		'B'
A.	Loop of Henle	5.	Plays a role in concentration of urine
B.	Renal artery	1.	Carries blood to the kidney
C.	Proximal convoluted tubule	2.	Area where a considerable amount of reabsorption takes place
D.	Glomerulus	4.	Filtration of blood
E.	Distal convoluted tubule	3.	Main area of K^+, H^+ secretion

Final Answer: (1)

179. Ans (4)

Solution:

	Column-I		Column-II
(a)	Pituitary gland	(iii)	Diabetes insipidus
(b)	Thyroid gland	(i)	Grave's disease
(c)	Adrenal gland	(iv)	Addison's disease
(d)	Pancreas	(ii)	Diabetes mellitus

Final Answer: (4)

180. Ans (2)

Solution: Actin and Myosin filaments do not shorten but rather slide pass each other

Final Answer: (2)